

UNIVERSIDAD FRANCISCO DE VITORIA

ESCUELA POLITÉCNICA SUPERIOR

COMPUTER SCIENCE DEGREE

FINAL DEGREE PROJECT

RESEARCH TRACK

Sign Lenguage Recognition System: Hand Detection and Word Classification using Deep Learning

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CALIFICACIÓN DEL PROYECTO DE FIN DE GRADO

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Lugar y fecha: Pozuelo de Alarcón, a ____ de _____ de 2026

Agradecimientos

Aquí se incluirían los agradecimientos personales y profesionales. No olvidarse de agradecer la ayuda recibida, especialmente si se ha realizado el proyecto gracias a una beca, contrato o algún tipo de subvención o apoyo.

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Resumen

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Palabras claves: reconocimiento de lengua de signos, detección de manos, aprendizaje profundo, YOLO, detección de objetos, visión por computador, ASL

Abstract

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Keywords: sign language recognition, hand detection, deep learning, YOLO, object detection, computer vision, ASL

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Chapter 1

Introduction

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1.1 Problem description

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1.2 Outline

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Chapter 2

State of the Art

2.1 From Symbolic to Subsymbolic AI

Early Artificial Intelligence (AI) systems were symbolic: they represented knowledge using explicit rules and logic, and solved problems through handcrafted if/then decision trees. This paradigm achieved success in constrained domains, but it struggled with perception tasks such as vision and speech because the real world is noisy, ambiguous, and high-dimensional. The so-called "knowledge acquisition bottleneck" meant that as tasks grew in complexity, engineering and maintaining rule bases became impossible. In the context of sign language, symbolic systems could encode simple gesture prototypes but failed to capture the fine-grained variability across signers, viewpoints, and execution speeds.

The shift to subsymbolic AI reframed intelligence as pattern recognition. Instead of encoding expert rules, models learn distributed representations from data. This change was prompted by advances in statistical learning and neural networks, culminating in deep learning methods that learn hierarchical features directly from raw inputs [1]. Subsymbolic models are particularly effective in perception because they optimize representations for the task objective, allowing subtle visual patterns to emerge without explicit programming. In sign language recognition, subsymbolic models learn the visual semantics of hand shapes, motions, and temporal dynamics from examples rather than hand-coded descriptors.

From a theoretical standpoint, subsymbolic approaches replace symbolic inference with continuous function approximation. A deep network implements a parameterized mapping f_θ that transforms inputs into task-specific embeddings, optimized by minimizing a loss over data. This paradigm aligns with the "computational" and "algorithmic" levels of explanation in vision, where representation learning is the core challenge. By training end-to-end models on large datasets, modern systems discover invariances to lighting, background, and signer variability that are nearly impossible to enumerate symbolically.

The subsymbolic shift also enabled modular pipelines in which perceptual frontends (e.g., hand pose estimation) feed compact, structured features to temporal models. This decoupling is powerful for SLR: the vision module learns spatial representations, while the sequence model learns dynamics. The result is a scalable, data-driven framework that has largely replaced rule-based approaches in

the literature. Contemporary SLR therefore reflects the broader AI trajectory: from symbolic rules to learned representations and, increasingly, to architectures that encode inductive biases (e.g., convolutional and recurrent structures) while remaining fully trainable from data.

2.2 Feature Extraction and Pose Estimation

The first challenge in Sign Language Recognition is extracting meaningful spatial features from video frames. This field has evolved from processing raw pixels to extracting high-level geometric structures.

2.2.1 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) represent the foundational technology for modern computer vision [1]. Unlike fully connected networks that treat inputs as flat vectors, CNNs preserve the spatial structure of images through three main operations: convolution, non-linearity, and pooling. A typical convolutional layer computes a feature map $\mathbf{y}$ as:

$$\mathbf{y}_{i,j,k} = \sigma\left(\sum_{c=1}^{C}\sum_{u,v}\mathbf{w}_{u,v,c,k}\,\mathbf{x}_{i+u,j+v,c} + b_k\right) \tag{2.1}$$

where $\mathbf{x}$ is the input tensor, $\mathbf{w}$ are learned kernels (filters), and σ is a non-linearity (e.g., ReLU). This operation introduces an inductive bias toward *translation equivariance*, meaning the network can recognize features regardless of their position in the image.

Deep CNN architectures, such as AlexNet [2] or ResNet, stack many such layers to learn hierarchical representations. Initial layers typically behave as Gabor filters, detecting simple edges and textures. Deeper layers combine these low-level features to detect complex shapes, and finally, semantic objects like hands or faces. The use of pooling layers further enhances robustness by introducing local translation invariance, allowing the network to detect features even if they are slightly shifted or distorted.

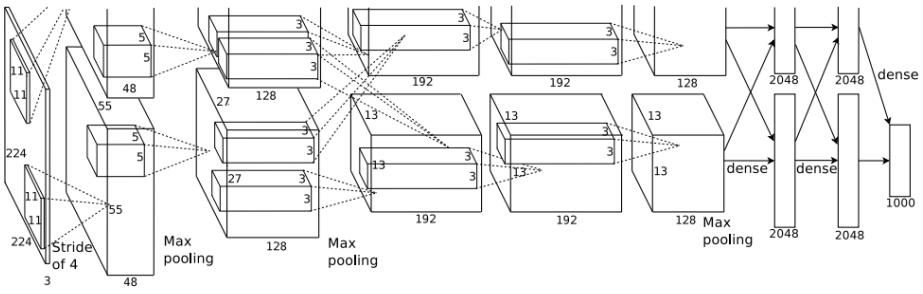


Figure 2.1: Architecture of AlexNet. Note the split processing paths originally designed for dual-GPU training. Source: Krizhevsky et al. [2].

2.2.2 From Pixels to Landmarks: Pose Estimation

While CNNs can process raw RGB pixels directly (pixel-based approach), this method is computationally expensive and sensitive to background noise and lighting conditions. To address this, *Pose Estimation* techniques leverage CNNs to regress the coordinates of anatomical keypoints, transforming the visual problem into a geometric one [3]. In practice, pose pipelines are commonly framed as either top-down (detect hands first, then regress keypoints) or bottom-up (predict keypoints first, then group them into hands). For SLR, the top-down approach is especially attractive because it focuses computation on the hand region and naturally supports frame-to-frame tracking.

MediaPipe Hands [4] is a seminal system in this domain. It employs a two-stage pipeline to achieve real-time performance:

1. **Palm Detector:** A lightweight single-shot detector operates on the full image to locate the hand palm. This step is crucial because detecting a rigid object like a palm is simpler than detecting articulating fingers.
2. **Landmark Regressor:** A high-capacity CNN processes the cropped hand region to predict the 3D coordinates of 21 skeletal joints.

This architecture decouples detection from regression, allowing the system to track hands efficiently by re-using the crop from the previous frame. For SLR, this provides a compact, invariant representation of the hand shape, filtering out irrelevant visual information such as background clutter or skin tone variations. The landmark representation also enables geometric normalization (e.g., translation and scale alignment) and the derivation of higher-level features such as bone vectors, joint angles, and inter-finger distances, which often yield more stable temporal dynamics than raw pixels.

Pose-based pipelines frequently integrate additional cues beyond the hands. Upper-body or face landmarks can capture non-manual signals (head pose, facial expression) that disambiguate signs or encode grammatical structure. Even when only hand keypoints are used, modern systems typically propagate per-joint confidence scores to handle occlusions and motion blur, masking unreliable points rather than forcing the sequence model to fit noise. These design choices make the pose representation more robust to real-world variability in signer appearance, speed, and viewpoint.

2.3 Sequence Modeling

Since sign language is inherently temporal, recognizing signs requires modeling the evolution of features over time. Standard feedforward networks cannot handle this task because they accept fixed-size inputs and have no memory of previous states. This limitation necessitates the use of Recurrent Neural Networks (RNNs).

2.3.1 Recurrent Neural Networks (RNNs)

RNNs introduced the concept of "memory" in neural networks by maintaining a hidden state $\mathbf{h}_t$ that is updated at each time step based on the current input $\mathbf{x}_t$ and the previous state $\mathbf{h}_{t-1}$:

$$\mathbf{h}_t = \phi(\mathbf{W}_{xh}\mathbf{x}_t + \mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{b}_h) \tag{2.2}$$

This recurrent connection allows the network to process sequences of variable length, sharing the same parameters (weights) across all time steps. Training is performed using *Backpropagation Through Time* (BPTT), which unrolls the network graph for the duration of the sequence.

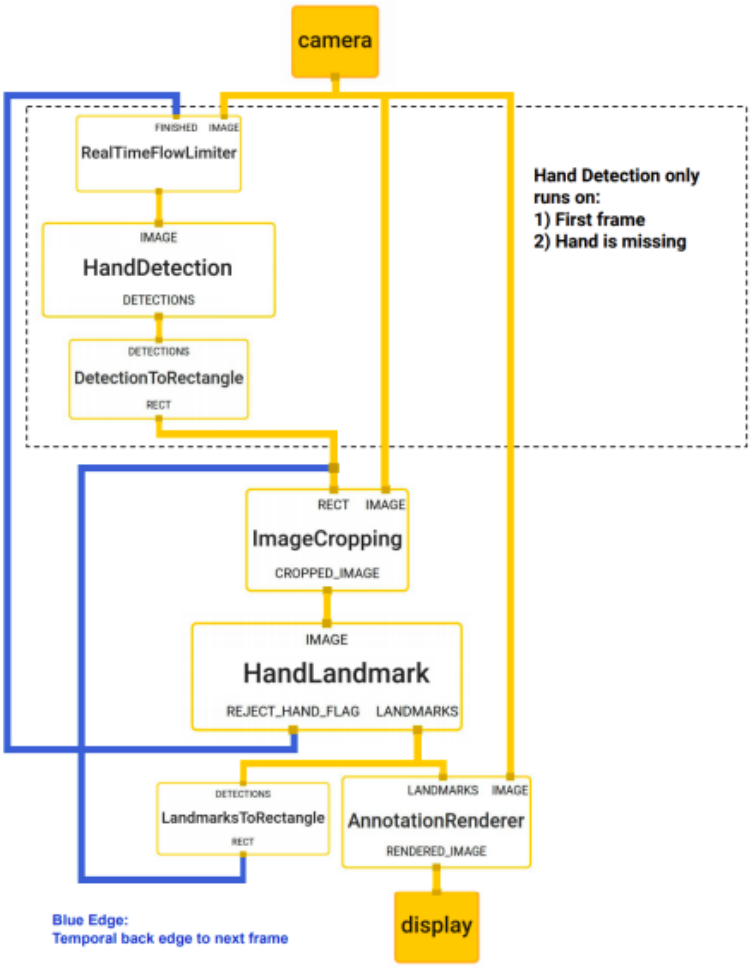


Figure 2.2: MediaPipe Hands pipeline showing the interaction between the detection and landmark models. Source: Zhang et al. [4].

However, RNNs suffer from a critical flaw known as the *vanishing gradient problem*. During BPTT, gradients are calculated by the chain rule, which involves repeated multiplication of the weight matrix. If the eigenvalues of this matrix are smaller than 1, the gradients decay exponentially as they propagate back to earlier time steps. Consequently, the network fails to learn long-term dependencies, effectively "forgetting" the beginning of a sign by the time it reaches the end.

2.3.2 Long Short-Term Memory (LSTM)

To overcome the vanishing gradient problem, Hochreiter and Schmidhuber [5] proposed the Long Short-Term Memory (LSTM) network. The core innovation of the LSTM is the separation of the

memory into two distinct vectors: the *Hidden State* ($\mathbf{h}_t$), which carries short-term information, and the *Cell State* ($\mathbf{C}_t$), which acts as a "highway" for information flow.

The Cell State runs down the entire chain with only minor linear interactions, making it easy for information to flow unchanged. This flow is regulated by three distinct "gates" composed of sigmoid neural network layers:

- **Forget Gate:** Decides what information to discard from the cell state (outputs numbers between 0 and 1).
- **Input Gate:** Decides what new values will be updated in the cell state.
- **Output Gate:** Controls what parts of the cell state are output to the hidden state.

This mechanism, known as the *Constant Error Carousel* (CEC), protects the error signal from decaying, enabling LSTMs to bridge time lags of more than 1000 steps. While highly effective, the complexity of calculating four distinct linear layers for each time step results in high computational costs and parameter counts. The internal structure of an LSTM cell is shown in **Figure 2.3(a)**.

2.3.3 Gated Recurrent Units (GRU)

Gated Recurrent Units (GRUs), proposed by Chung et al. [6], represent a modern simplification of the LSTM architecture. GRUs address the redundancy in LSTMs by removing the separate cell state and merging the input and forget gates into a single **Update Gate** ($\mathbf{z}_t$). They also introduce a **Reset Gate** ($\mathbf{r}_t$).

The mathematical formulation for a GRU at time t is:

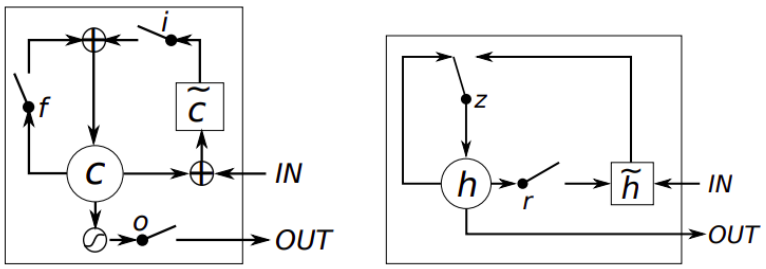
$$\mathbf{z}_t = \sigma(\mathbf{W}_z \mathbf{x}_t + \mathbf{U}_z \mathbf{h}_{t-1} + \mathbf{b}_z) \tag{2.3}$$

$$\mathbf{r}_t = \sigma(\mathbf{W}_r \mathbf{x}_t + \mathbf{U}_r \mathbf{h}_{t-1} + \mathbf{b}_r) \tag{2.4}$$

The candidate hidden state $\tilde{\mathbf{h}}_t$ uses the reset gate to drop irrelevant past information, while the final hidden state is a linear interpolation controlled by the update gate:

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t \tag{2.5}$$

This interpolation is the key to GRU's efficiency: the network can decide to copy the previous state $\mathbf{h}_{t-1}$ directly (when $\mathbf{z}_t \approx 0$), preserving long-term memory without the complex machinery of the LSTM. Illustrated in **Figure 2.3(b)**, GRUs typically match LSTM performance with fewer parameters, making them less prone to overfitting on datasets with limited samples.



(a) Long Short-Term Memory (b) Gated Recurrent Unit

Figure 2.3: Comparison of Recurrent Units. (a) Long Short-Term Memory (LSTM) with three gates. (b) Gated Recurrent Unit (GRU) with two gates. Source: Chung et al. [6].

2.4 Attention Mechanisms

Attention mechanisms emerged as a response to the "information bottleneck" in fixed-length sequence encodings, first popularized in neural machine translation by Bahdanau et al. [7]. In traditional Recurrent architectures, the entire input sequence must be compressed into a single context vector $\mathbf{v}$. For long sequences, this compression results in loss of detail, as the vector size is fixed regardless of the input length.

Attention solves this by allowing the model to dynamically search for parts of the source sequence that are relevant to the current prediction. Instead of a static vector, Attention computes a *Context Vector* as a weighted combination of all encoder states $\{\mathbf{h}_t\}_{t=1}^{T}$.

Formally, the system learns an alignment model (or scoring function) $a(\mathbf{s}, \mathbf{h}_t)$ that measures how well the inputs around position t and the output at position i match. These scores are normalized using a Softmax function to produce attention weights α_t:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)}, \quad e_t = a(\mathbf{s}, \mathbf{h}_t) \tag{2.6}$$

The context vector $\mathbf{c}$ is then derived as:

$$\mathbf{c} = \sum_{t=1}^{T} \alpha_t \mathbf{h}_t \tag{2.7}$$

In the context of SLR, temporal attention acts as a learned filter. Sign language videos often contain "non-discriminative" frames, such as the hands returning to a resting position or preparatory movements. By assigning near-zero weights ($\alpha_t \approx 0$) to these frames, the model effectively ignores the noise and focuses its computational capacity on the "key frames" where the semantic meaning of the sign is contained. This improves robustness to variable execution speeds and intra-class variability.

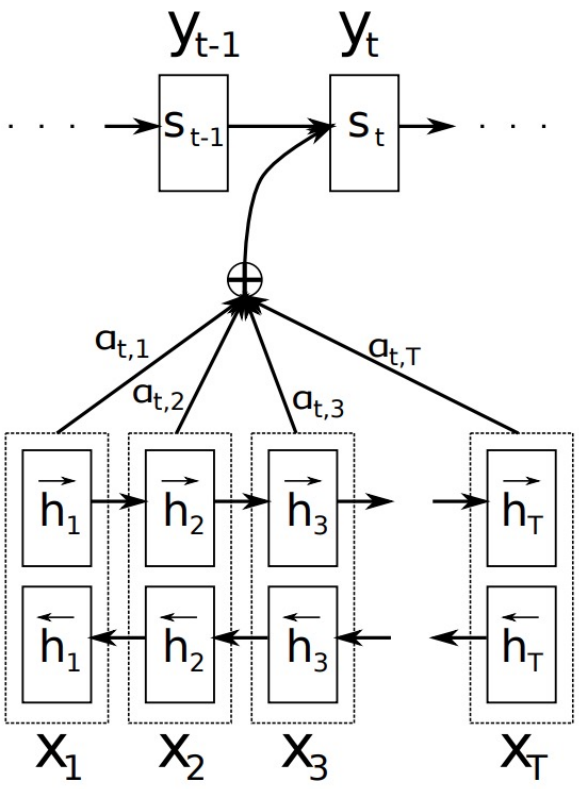


Figure 2.4: Neural Machine Translation architecture with Attention mechanism. The context vector is a weighted sum of encoder annotations. Source: Bahdanau et al. [7].

Chapter 3

Objectives

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3.1 Main objective

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3.2 Specific objectives

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3.3 Validation methods

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Chapter 4

Methodology and Work Plan Overview

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4.1 Materials and Resources

4.2 Methods

4.3 Methodology

4.4 Work Plan

4.5 Constraints and Limitations

Chapter 5

Technical Development

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5.1 System design

Figure 5.1 shows the class diagram.

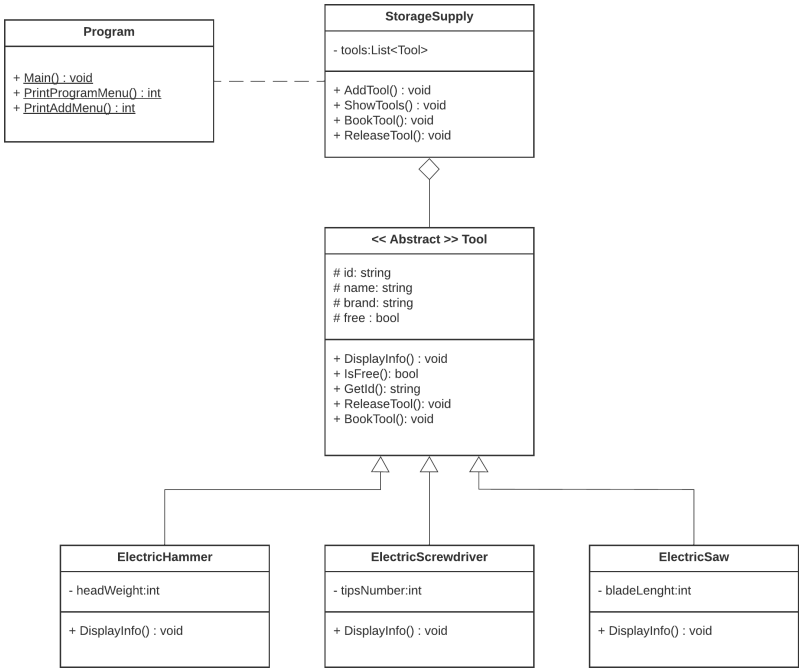


Figure 5.1: Class diagram example.

5.1.1 Description of functional features

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5.1.2 Operational Environment

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5.2 Requirements

5.2.1 Description of requirement attributes

For the textual description of the different identified requirements, a set of attributes has been selected to describe each requirement. Below is a description of the meaning of each attribute used:

- **Code**: A unique abbreviated identifier for the requirement. It is constructed using the requirement type code followed by a hyphen and three digits. Requirements are divided into functional and non-functional requirements. The codes used are **FR** for Functional Requirements and **NFR** for Non-Functional Requirements. For example: `FR-001`.
- **Name**: Extended identification of the requirement.
- **Description**: A basic description of the identified requirement.
- **Source**: Indicates the source through which the requirement was identified. This value usually corresponds to one or more use case codes.
- **Necessity**: Determines the degree of implementation required for the requirement. The possible values are:
  - **Essential**: The requirement must be implemented.
  - **Desirable**: It is preferable to implement the requirement, but it is not mandatory.
  - **Optional**: The requirement may be implemented, but it is neither important nor mandatory.
- **Priority**: Defines the importance of the requirement, allowing the order in which requirements are included in the design and implementation process to be established. The possible values are:
  - **High**: The requirement must be implemented in the early stages of development.

  - **Medium**: The requirement should be implemented once high-priority requirements have been completed.
  - **Low**: The requirement should be implemented in the final stages of development. These requirements do not affect the correct functioning of the system.
- **Stability**: Defines the stability of the requirement throughout the software life cycle. This indicates whether the requirement may be modified during the life cycle. The possible values are:
  - **Stable**: The requirement will not change during the system life cycle.
  - **Unstable**: The requirement may change during the system life cycle.
- **Verifiability**: Defines the degree to which a requirement can be verified, that is, the extent to which it is possible to confirm that the requirement has been implemented in the developed system. The possible values are:
  - **High**: It is possible to verify that the requirement has been implemented. These requirements usually correspond to core system functionalities.
  - **Medium**: It is possible to verify that the requirement has been implemented, but it requires complex testing or inspection of the source code.
  - **Low**: It is difficult, or in some cases impossible, to verify whether the requirement has been implemented.

System requirement			
ID	111	**Source**	111
Name	111		
Description	111		
Necessity	111	**Priority**	111
Stability	111	**Verifiability**	111

Table 5.1: Functional requirement 1

5.3 Datasets

This project spans two recognition tasks with fundamentally different data requirements: hand localization from still images, and word-level sign language recognition from video. As a consequence, five distinct datasets are employed across the full experimental pipeline. This section formalizes the properties of each dataset by first defining a common set of descriptive attributes, then presenting a unified comparative table, and finally providing individual descriptions of each dataset in context.

5.3.1 Dataset Attribute Definitions

To ensure a consistent and unambiguous characterization of every dataset, the following attributes are used in Table 5.2. Each attribute captures a distinct dimension relevant to dataset selection, reproducibility, and the interpretation of experimental results:

- **ID**: A unique alphanumeric identifier assigned to each dataset within this thesis. The prefix `DS-` followed by a two-digit index allows unambiguous cross-referencing in subsequent chapters when reporting experimental configurations and results.
- **Name**: The official or widely-used name of the dataset as it appears in the literature or on its hosting platform.
- **Year**: The year of original publication or first public release. This contextualizes the dataset within the chronological development of the field and helps evaluate whether it reflects the current state of the art in data collection methodology.
- **Task**: The specific sub-task for which the dataset was designed. Three task categories appear in this project:
  - **Hand Detection**: the dataset provides annotated bounding boxes or segmentation masks marking the location of hands in images.
  - **Letter Recognition**: the dataset provides labeled static images of individual American Sign Language (ASL) letter signs.
  - **Word Recognition**: the dataset provides labeled video clips of complete ASL word signs, requiring temporal modeling.
- **Samples**: The total number of annotated data instances. For image datasets, this equals the number of images; for video datasets, it equals the number of individual video clips. Where a subset of the full dataset was used in this project, this is noted in the individual descriptions.
- **Classes**: The number of distinct output categories. For hand detection datasets this is always 1 (the single *hand* class); for sign language datasets it corresponds to vocabulary size — the number of distinct letters or word glosses.
- **Format**: The storage format and annotation type of the raw data. This attribute specifies whether instances are static images or video clips, the image resolution where relevant, and the form of ground-truth annotation (e.g., bounding box coordinates in XML, pixel-level segmentation masks, image class labels, or video-level class labels).
- **Input Representation**: Whether the models trained on this dataset consume *pixel-based* data (raw RGB or grayscale image/video frames) or *landmark-based* data (structured 3D keypoint coordinates extracted by a pose estimator such as MediaPipe). Pixel-based inputs preserve full visual detail but are high-dimensional and sensitive to background, lighting, and signer appearance. Landmark-based inputs are compact, viewpoint-invariant, and more robust to visual variability, at the cost of depending on a reliable keypoint extractor.

5.3.2 Comparative Overview

Table 5.2 provides a high-level comparison of the five datasets used in this project. Datasets DS-01 and DS-02 support the hand localization pipeline described in later experimental sections. Datasets DS-03 and DS-04 were used in the initial letter-level classification phase of the project, which was subsequently superseded by the word recognition approach. Dataset DS-05 is the primary benchmark for all word-level recognition experiments and constitutes the main evaluation resource throughout the thesis.

Table 5.2: Comparative overview of all datasets used in this project.

ID	Name	Year	Task	Samples	Classes	Format
DS-01	EgoHands	2015	Hand Detection	4,800	1	Images + Seg. Masks
DS-02	Kaggle Hand Dataset	2023	Hand Detection	∼4,200	1	Images + BBox (XML)
DS-03	Sign Language MNIST	2017	Letter Recog.	34,627	24	28×28 Grayscale CSV
DS-04	ASL Alphabet	2018	Letter Recog.	87,000	29	200×200 RGB JPG
DS-05	WLASL	2020	Word Recog.	21,083	2,000	Video Clips (RGB)

5.3.3 Hand Detection Datasets

Datasets DS-01 and DS-02 are used jointly to train the hand localization module based on YOLOv8. A central design motivation for combining two datasets is improved generalization: a detector trained exclusively on egocentric footage tends to fail when the camera is positioned at eye level or beyond arm's reach, while a detector trained only on third-person images may struggle with first-person perspectives. By merging the two sources, the training corpus covers a significantly wider range of viewpoints, backgrounds, and hand appearances. After merging, duplicate removal, and a stratified train/validation/test split, the combined training set comprises approximately 5,223 annotated images.

DS-01 — EgoHands [8]

EgoHands is an egocentric hand detection dataset published by Indiana University in 2015 as part of research on activity recognition in close-proximity social interactions. It consists of approximately 4,800 labeled frames extracted from 48 video recordings captured from a GoPro camera mounted on the chest of an observer watching another person. Ground-truth annotations are stored as pixel-level segmentation polygons in MATLAB `.mat` files; during preprocessing, these polygons were converted to axis-aligned bounding boxes in YOLO format. The dataset covers diverse interaction scenarios — including card games, chess, and jigsaw puzzles — with two hands simultaneously visible in most frames, providing challenging overlap and occlusion cases. Because all images are taken from a first-person perspective, the visual distribution of EgoHands is inherently biased toward the observer's own hand. This limitation motivates the inclusion of DS-02 to extend viewpoint diversity. The input representation is pixel-based: models consume raw RGB crops.

DS-02 — Kaggle Hand Dataset [9]

The Kaggle Hand Dataset is a collection of annotated hand images published on the Kaggle platform, providing multi-viewpoint coverage that is complementary to the egocentric nature of DS-01. Images depict hands in third-person perspectives with diverse backgrounds, lighting conditions, skin tones, and partial occlusions. Ground-truth bounding boxes are provided in Pascal VOC XML format; during preprocessing, coordinates were normalized to the YOLO format used by the detection pipeline. The dataset contains approximately 4,200 labeled images. The joint training corpus formed by DS-01 and DS-02 yielded a YOLOv8n model achieving approximately 98% mAP50 and 87%

Mean IoU on the held-out test split, demonstrating the effectiveness of the multi-source approach. The input representation is pixel-based.

5.3.4 Letter Recognition Datasets

Datasets DS-03 and DS-04 were used during the initial phase of the project, when the recognition target was the static letter-level ASL alphabet. Both datasets cover 24 of the 26 ASL letters; the letters *J* and *Z* are excluded in both cases because they require hand motion and cannot be represented by a single static image. The two sources were merged and unified into a single training corpus for letter classification experiments. Although this phase was later superseded by the more ambitious word-level recognition approach, these datasets remain relevant because they established the viability of data-driven letter classification and informed the design of the preprocessing pipeline. The input representation for both datasets is pixel-based.

DS-03 — Sign Language MNIST [10]

Sign Language MNIST mirrors the format of the canonical MNIST handwritten digit benchmark. It contains 27,455 training images and 7,172 test images — 34,627 samples in total — each representing one of 24 static ASL letters as a 28×28 grayscale image. Data is distributed in CSV format, with one row per image where each column corresponds to a pixel intensity value in the range $[0, 255]$, plus a label column. The low resolution and grayscale format make this dataset extremely lightweight and straightforward to preprocess, but inherently limit the discriminative information available to the model, as fine-grained finger articulation details are largely lost. The dataset was originally constructed from ASL alphabet photographs rescaled and converted to the MNIST-style format, and is hosted on Kaggle.

DS-04 — ASL Alphabet [11]

The ASL Alphabet dataset provides a substantially richer visual input than DS-03. It contains approximately 87,000 images organized into 29 classes: the 24 static ASL letters, the letter *space*, and two additional non-letter classes (*del*, *nothing*). Each class is represented by exactly 3,000 training images captured in a controlled environment at a resolution of 200×200 pixels in full RGB color. The higher resolution and color information enable more discriminative feature learning, particularly for hand texture and illumination cues. For integration with DS-03 into a unified corpus, images were converted to grayscale and resized to a common format, yielding a combined training set of approximately 79,000 images and 7,000 test images covering the 24 shared letter classes.

5.3.5 Word Recognition Dataset

DS-05 — WLASL [12]

The Word-Level American Sign Language (WLASL) dataset, published by Li et al. in 2020, is the primary data resource for all word-level recognition experiments in this thesis and one of the largest publicly available ASL word recognition benchmarks to date. WLASL contains 21,083 video instances covering 2,000 ASL glosses (words). Videos were collected from multiple online platforms — including YouTube channels maintained by ASL educators, online courses, and signing dictionaries — and manually verified by qualified ASL experts to ensure correctness of sign production. Each clip shows a single signer performing one complete sign, typically against a plain or low-clutter background, with durations ranging from approximately one to four seconds.

The dataset is partitioned into four progressively larger evaluation subsets: WLASL100 (100 glosses), WLASL300 (300 glosses), WLASL1000 (1,000 glosses), and WLASL2000 (the full 2,000 glosses). This structure enables systematic benchmarking of how model performance scales with vocabulary size. A practical challenge is that WLASL relies on external video URLs; a fraction of clips have become unavailable over time due to deletion or access restriction, so only successfully downloaded videos can be used.

In this project, raw RGB frames are not used as model input. Instead, MediaPipe Hands [4] is applied to each frame to extract 21 three-dimensional hand keypoints, yielding a compact landmark vector of 63 scalar features per frame (21 joints $\times$ 3 coordinates). This representation reduces input dimensionality by a factor of approximately 2,389 compared to raw 224×224 RGB frames, removes background and appearance noise, and provides geometric invariance to signer appearance. The effective detection rate of MediaPipe on WLASL clips is approximately 69%; frames where no hand is detected are linearly interpolated from adjacent detected frames to maintain sequence continuity. Each video is resampled to a fixed sequence length of 64 frames before being fed to the model.

For the experiments described in this thesis, the following WLASL subsets were extracted and used:

- **WLASL30**: The 30 most frequent glosses, used for initial architecture exploration and model selection. Split: 242 training / 56 validation / 34 test videos.
- **WLASL100**: The 100 most frequent glosses, used for K-Fold cross-validation and robust evaluation. Split: 741 training / 162 validation / 99 test videos.
- **WLASL300, WLASL1000, WLASL2000**: Larger subsets targeted for progressive scaling experiments in later stages of the project, where the goal is to assess model accuracy and generalization at increasing vocabulary sizes.

Chapter 6

Experiments

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Chapter 7

Ethical Implications and Social Impact

7.1 AI Based Systems Ethical Impact

An ethical analysis in Artificial Intelligence is a process which consists in examining the implications, concerns and challenges associated with the development and deployment of these systems. With the recent advancements, AI is starting to integrate in applications which can be considered critical to society, and that is why their technical improvement and evaluation should also be accompanied by an ethics review. This ethical impact review on AI systems is based on 4 general pillars:

- **Human Rights and Fairness:** it is necessary to prevent biases and discrimination against minority groups to ensure fairness in treatment.
- **Accountability:** it is crucial to establish who is responsible for the decisions the AI systems make, whether that is the development team, the users, the company, or even the system itself.
- **Privacy and Security:** it must be ensured that user data is protected and that the system does not infringe any confidentiality of the users.
- **Transparency:** it is indispensable to provide clarity on how the system is taking decisions and what they are based on, meaning black box systems should be avoided.

7.2 Ethical Evaluation Framework: Google's AI Principles

Several organizations have developed frameworks to standardize the ethical evaluation of AI systems. For the evaluation of this project, the framework adopted is **Google's AI Principles** [13], originally published in 2018 by Google CEO Sundar Pichai. This framework has been updated and detailed through annual progress reports, and the version used as the primary reference for this evaluation is the **2023 Progress Update** [14], which represents the fifth edition of these reports and provides a comprehensive description of how the principles are applied in practice across different types of AI systems.

The reason for choosing this specific framework is that it provides a clear, well structured set of seven principles that are practical enough to apply directly to a research project. Unlike other frameworks that tend to be more abstract or policy oriented, Google's AI Principles are formulated as concrete objectives that can be evaluated with quantifiable metrics. Furthermore, Google has been one of the pioneers in publishing annual progress reports on how these principles are applied in real products and services, which gives the framework a level of maturity and real world validation that makes it suitable for academic evaluation. The 2023 report, in particular, demonstrates how these principles can be put into practice across different types of AI systems, including generative AI, computer vision, and speech recognition, which aligns well with the nature of this project.

The seven principles that make up this framework are:

1. **Be socially beneficial:** The AI system should aim at solving a real societal problem and should do it effectively.
2. **Avoid creating or reinforcing unfair bias:** Steps should be taken to ensure that the AI system is fair and inclusive for everyone, avoiding discrimination.
3. **Be built and tested for safety:** Safety should be a priority in how the system is designed and developed.
4. **Be accountable to people:** The system should provide transparency and allow users to have some control over it.
5. **Incorporate privacy design principles:** The AI should be designed in a way that protects user data and privacy.
6. **Uphold high standards of scientific excellence:** The development of the AI should be based on rigorous research and sound scientific principles.
7. **Be made available for uses that accord with these principles:** The intended use cases for the AI should be clear, making sure it is used properly and aligns with these values.

In the following sections, the ASL recognition system developed in this project is evaluated against each of these seven principles. For each principle, a set of quantitative metrics is defined to assess to what degree the project fulfills that principle. Each metric is scored on a simplified scale of **0**, **5** or **10**, where 0 means the principle is not addressed at all, 5 means it is partially addressed, and 10 means it is fully addressed. This scoring system keeps the evaluation straightforward while still providing enough granularity to distinguish between different levels of compliance. The metrics may or may not come directly from experimental results; some are derived from design decisions, documentation practices, or the nature of the data used. After evaluating all principles individually, a summary table is presented with the scores for each metric and the average score per principle, which provides an overall view of the ethical compliance of the project.

7.3 Project Analysis Against Google's AI Principles

7.3.1 Principle 1: Be Socially Beneficial

The primary goal of this project is to develop an AI system capable of recognizing American Sign Language (ASL) words from video sequences. Sign language is the main means of communication for millions of deaf and hard of hearing individuals worldwide, and yet there is still a significant communication barrier between the deaf community and hearing people who do not know sign

language. An automatic ASL recognition system can serve as a bridge to reduce this gap, potentially enabling real time translation in educational, medical and everyday settings.

To evaluate how well this project fulfills the social benefit principle, the following metrics are defined:

- **M1.1 — Addresses a real societal need:** The system targets a genuine accessibility problem. According to the World Health Organization, over 430 million people worldwide have disabling hearing loss. Providing tools that facilitate communication for this community represents a clear social benefit. *Score: 10.*
- **M1.2 — Effectiveness of the solution:** The best model achieved 63.9% Top-1 accuracy (K-Fold validated) and 86.5% Top-5 accuracy on a 30-class vocabulary, meeting the baseline accuracy targets established for the project. However, accuracy degrades when scaling to larger vocabularies, reaching 44.7% at 100 classes and 28.6% at 300 classes. The system shows promise as a research prototype but has not yet reached the level of reliability required for real world deployment. *Score: 5.*
- **M1.3 — Accessibility of the technology:** The system is built entirely on open source tools (Python, PyTorch, MediaPipe) and uses publicly available datasets (WLASL). The research is documented in a university thesis, making the methodology accessible to other researchers. Additionally, the project was designed with the intention of running the recognition pipeline within a mobile application, which would make it accessible to end users regardless of their technical background. *Score: 10.*

7.3.2 Principle 2: Avoid Creating or Reinforcing Unfair Bias

Bias in AI systems is a well known problem, and it is particularly relevant in sign language recognition because different signers may have different hand sizes, skin tones, signing speeds, and styles. If the training data is not diverse enough, the model could perform significantly better for some demographics than for others. This concern applies not only to the word recognition phase but also to the earlier hand detection phase, which was trained on image datasets containing real photographs of people.

- **M2.1 — Dataset diversity (word recognition):** The WLASL dataset, used for word level recognition, contains videos from multiple signers with varying backgrounds. However, the dataset does not provide detailed demographic annotations (age distribution, skin tone, hand size, etc.), which makes it difficult to formally assess the diversity of the training data. On the other hand, the fact that the videos were collected from different YouTube sources means that there is likely some natural variation in the signers, even though this cannot be quantified. *Score: 5.*
- **M2.2 — Dataset diversity (hand detection and letter classification):** The hand detection models (YOLOv8n, SSDLite-MobileNetV3, EfficientDet-D0) were trained on a combination of four datasets: EgoHands (∼4,800 egocentric images from Indiana University), Oxford/Kaggle Hand Dataset (∼13,000 images), HaGRID (∼30,000 gesture images with bounding boxes), and TV-Hand (∼5,000 images). Additionally, the initial letter classification experiments used the Sign Language MNIST (∼7,000 grayscale images) and the ASL Alphabet Dataset (∼72,000 RGB images). These datasets have varying levels of diversity: EgoHands is limited to a university context with an egocentric perspective, while HaGRID and the Oxford set offer more variety. None of them provide explicit demographic metadata, which means

that potential biases in skin tone or hand morphology representation cannot be formally ruled out. *Score: 5.*

- **M2.3 — Input representation fairness:** For the word recognition phase, the system uses MediaPipe hand landmarks (21 keypoints $\times$ 3 coordinates = 63 features per frame) instead of raw pixel data. This means that the model operates on geometric joint positions rather than visual appearance, which makes it inherently less sensitive to factors like skin color, lighting conditions or background. This significantly reduces the risk of appearance based bias in the recognition stage. However, it should be noted that the hand detection stage and the MediaPipe extraction itself do operate on raw images, so bias could still be introduced at the detection level. *Score: 10.*
- **M2.4 — Per-class performance analysis:** Per class accuracy analysis was conducted across all experiments, identifying that some classes like "what" (8.3%) and "cool" (12.5%) had significantly lower accuracy than others like "orange" (100%) and "table" (100%). These disparities are mostly explained by visual similarity between certain signs and the limited number of training samples for some classes, rather than by demographic bias. While the disparities were documented and analyzed, specific corrective measures such as class weighted loss or oversampling were not implemented in this iteration, as the focus was on establishing baseline performance across progressive vocabulary sizes. The system does log per class metrics as part of training, which would facilitate targeted improvements in future iterations. *Score: 5.*

7.3.3 Principle 3: Be Built and Tested for Safety

Safety in this context refers to ensuring that the AI system operates reliably and does not produce harmful or misleading outputs. For a sign language recognition system, a misclassification could lead to a misunderstanding, which in a medical or legal context could have serious consequences.

- **M3.1 — Robust validation methodology:** Instead of relying on a single train-test split, Stratified 5-Fold Cross-Validation was implemented, which provides a more reliable and statistically meaningful estimate of model performance. In this approach, the dataset is divided into 5 non overlapping folds, and the model is trained and evaluated 5 times, each time using a different fold as the test set and the remaining 4 folds for training. The stratification ensures that the class distribution is preserved in each fold. This yielded a mean Top-1 accuracy of 63.9% $\pm$ 4.5% and a mean Top-5 accuracy of 86.5% $\pm$ 2.0% across the 5 folds. The use of Test-Time Augmentation (TTA) was explicitly avoided because it produces artificially inflated results that do not reflect the true performance of the model. *Score: 10.*
- **M3.2 — Scalability testing:** Progressive training was performed across multiple WLASL subsets (30, 100, 300, 500, 700, 900, 1000, 1200, 1500 classes) to understand how the model degrades as the task complexity increases. This allowed the identification of instability points (above 700 classes for GRU+Attention) and the documentation of the limitations of the current approach. This type of progressive stress testing is an important safety practice because it establishes the boundaries within which the system can be considered reliable. *Score: 10.*
- **M3.3 — Hand detection reliability:** The hand detection stage, based on YOLOv8n, achieved approximately 98% mAP50 on the combined test set. This high detection rate is important for overall system safety, as a missed hand detection would prevent the recognition pipeline from even attempting classification. The detection model was validated on images from all four

source datasets to ensure consistent performance across different visual contexts. *Score: 10.*

- **M3.4 — Failure mode documentation:** Per class confusion matrices were documented and problematic word pairs that the model confuses were identified. However, a confidence threshold mechanism that would allow the system to abstain from prediction when uncertain has not been implemented yet. Such a mechanism would be important in a production deployment to avoid presenting unreliable predictions to the user, and it is considered as a future improvement. *Score: 5.*

7.3.4 Principle 4: Be Accountable to People

Accountability means that the system should be transparent in its operation and that users should have some level of understanding and control over its behavior.

- **M4.1 — Model interpretability:** The GRU + Temporal Attention architecture includes an attention mechanism that produces weights over the temporal sequence of input frames. These attention weights indicate which frames the model considers most relevant for making its prediction, which provides a degree of interpretability. Unlike a purely opaque model where the decision process is entirely hidden, the attention weights can be inspected and visualized to understand why a particular prediction was made. This is an advantage over architectures that do not include any form of explainability. *Score: 10.*
- **M4.2 — Top-K predictions:** The system provides Top-5 predictions alongside Top-1, giving users multiple possible interpretations rather than a single definitive answer. At 30 classes, the Top-5 accuracy is 86.5%, meaning that the correct word is almost always among the top 5 suggestions. This allows the end user to make the final decision from a shortlist, rather than being forced to accept or reject a single prediction. *Score: 10.*
- **M4.3 — Reproducibility:** All experiments are documented in Jupyter notebooks with fixed random seeds, and the training configurations, hyperparameters and dataset splits are recorded. However, some WLASL videos come from YouTube and may become unavailable over time, which threatens long term reproducibility. Additionally, the YouTube videos used in WLASL were collected without explicit consent from the signers for use in machine learning research, which is an ethical concern regarding data provenance that affects the ability to share and reproduce the dataset freely. *Score: 5.*

7.3.5 Principle 5: Incorporate Privacy Design Principles

Privacy is an important consideration in any system that processes video data of people. Sign language recognition inherently requires capturing video of individuals performing gestures, which raises questions about data protection. This project involves multiple stages with different privacy characteristics: the hand detection training phase used real images of people, the word recognition training phase used abstract landmark representations, and the inference pipeline processes raw video frames sent from a mobile device.

- **M5.1 — Data minimization through landmarks (training phase):** One of the most significant privacy advantages of the word recognition approach is the use of MediaPipe landmarks instead of raw video. The landmark extraction process converts each video frame into 63 numerical values (21 keypoints $\times$ 3 coordinates), completely discarding the original visual

information. Once landmarks are extracted, the original video is no longer needed for training or inference of the GRU/LSTM models. This means that the processed data cannot be used to reconstruct the person's face, body, or surroundings, which is a strong form of data minimization. *Score: 10.*

- **M5.2 — Privacy in the hand detection phase:** The hand detection models were trained on datasets containing real photographs of people (EgoHands, Oxford/Kaggle, HaGRID, TV-Hand), totaling approximately 53,000 images. EgoHands in particular contains frames from video recordings of people in everyday situations (eating, playing, etc.). Similarly, the letter classification experiments used approximately 79,000 real hand images from Sign Language MNIST and the ASL Alphabet Dataset. These images were processed during training but are not stored or distributed as part of the final system. Nevertheless, the fact that real images of identifiable individuals were used in the training pipeline is a privacy consideration that should be acknowledged. These datasets are publicly available for research purposes, but no independent privacy audit was conducted on their provenance. *Score: 5.*
- **M5.3 — Privacy in the inference pipeline:** The current server implementation (FastAPI) receives full video frames (416×416 RGB) from the mobile application, processes them for hand detection, and stores both the raw frames and cropped hand regions as `.npy` and `.png` files in session directories on the server. This means that identifiable visual data travels over the network and is persisted on disk without encryption or automatic expiration. This is the weakest point of the system from a privacy perspective. In a production deployment, the landmark extraction should ideally be performed on the client device so that only abstract landmark data is transmitted to the server, or alternatively, the server should implement encryption and automatic session data deletion. *Score: 0.*
- **M5.4 — K-Anonymity considerations:** K-Anonymity is a privacy model in which a dataset is considered k-anonymous if every combination of almost identifying attributes (such as age, gender, or location) corresponds to at least k individuals, making it impossible to single out any one person [15]. In this project, the WLASL dataset does not include identifying metadata about the signers (no names, ages, or locations are stored alongside the landmark data). Furthermore, the landmark representation itself acts as a form of generalization: the 63 dimensional feature vector cannot be easily traced back to a specific individual because different people performing the same sign produce similar landmark patterns. While a formal k-anonymity algorithm was not applied, the design of the landmark pipeline achieves a similar privacy preserving effect for the word recognition data. This does not apply, however, to the hand detection datasets or the server inference pipeline, where raw images are involved. *Score: 5.*

7.3.6 Principle 6: Uphold High Standards of Scientific Excellence

This principle evaluates whether the research and development process follows rigorous scientific methodology.

- **M6.1 — Systematic model comparison (word recognition):** A structured comparison of multiple architectures was conducted for the word recognition task: CNN+LSTM (pixel based), LSTM (landmark based), GRU with soft augmentation, GRU with enriched features, GRU with Temporal Attention, and a Transformer model. Each architecture was evaluated under the same conditions and datasets, allowing for a fair and controlled comparison. *Score: 10.*

- **M6.2 — Systematic model comparison (hand detection):** For the hand detection stage, three different object detection architectures were trained and compared: YOLOv8n, SSDLite320-MobileNetV3, and EfficientDet-D0. These models were evaluated on the same combined dataset using standard object detection metrics (mAP50, mAP50-95, inference time), providing a rigorous comparison across different architecture families ranging from one stage to two stage detectors. *Score: 10.*
- **M6.3 — Statistical validation:** The use of 5-Fold Stratified Cross-Validation with reported mean and standard deviation (63.9% $\pm$ 4.5%) provides a statistically meaningful evaluation. Additionally, Pearson correlation coefficients were computed to analyze the relationship between the number of classes and model performance during the scalability analysis, identifying quantitative trends in accuracy degradation. *Score: 10.*
- **M6.4 — Literature grounding:** The state of the art chapter of this thesis covers the theoretical progression from CNNs to RNNs, LSTMs, and GRUs, situating the work within the broader research landscape. The architectures chosen (GRU+Attention, LSTM, Transformer) are well established in the temporal sequence modeling literature. The landmark based approach using MediaPipe is also grounded in recent work on pose estimation for sign language recognition. *Score: 10.*
- **M6.5 — Honest reporting of limitations:** The thesis explicitly documents accuracy degradation at scale (from 63.9% at 30 classes to 11.2% at 1000 classes), MediaPipe detection failures ($\sim$69% frame detection rate), and the instability observed above 700 classes. Negative results and limitations are reported alongside positive findings, which is consistent with the principle of scientific integrity. *Score: 10.*

7.3.7 Principle 7: Be Made Available for Uses That Accord with These Principles

This principle asks whether the intended uses of the technology align with the ethical framework and whether potential misuse has been considered.

- **M7.1 — Intended use clarity:** The system is designed specifically as a research tool for ASL word recognition. The thesis clearly defines the scope of the project, the target application (assistive technology for the deaf community), and the current limitations. There is no ambiguity about the intended use. *Score: 10.*
- **M7.2 — Misuse potential assessment:** The landmark based approach for word recognition produces abstract numerical representations that have very limited potential for misuse. Unlike a system that processes raw video, the landmark features cannot be repurposed for surveillance, facial recognition, or tracking individuals. However, it should be noted that the hand detection component and the server inference pipeline do process real images, which could theoretically be repurposed if the server data were accessed by unauthorized parties. The current prototype does not implement access controls or data retention policies on the session storage, which is a gap that should be addressed before any deployment beyond the research context. *Score: 5.*
- **M7.3 — Open research contribution:** The methodology, results, and analysis are documented in a university thesis and the code is developed in documented Jupyter notebooks. This contributes to the open research community working on sign language recognition. However,

the trained models and datasets have not been published on a public repository at the time of writing. *Score: 5.*

7.4 Impact Scale of Ethical Aspects

Table 7.1 summarizes all the metrics defined for each of Google's AI Principles, along with their individual scores. The final column shows the average score per principle.

Principle	ID	Metric	Score	Avg.
P1: Socially Beneficial	M1.1	Addresses a real societal need	10	**8.3**
	M1.2	Effectiveness of the solution	5	
	M1.3	Accessibility of the technology	10	
P2: Avoid Unfair Bias	M2.1	Dataset diversity (word recognition)	5	**6.3**
	M2.2	Dataset diversity (detection & letters)	5	
	M2.3	Input representation fairness	10	
	M2.4	Per class performance analysis	5	
P3: Safety	M3.1	Robust validation methodology	10	**8.8**
	M3.2	Scalability testing	10	
	M3.3	Hand detection reliability	10	
	M3.4	Failure mode documentation	5	
P4: Account-ability	M4.1	Model interpretability	10	**8.3**
	M4.2	Top-K predictions	10	
	M4.3	Reproducibility	5	
P5: Privacy	M5.1	Data minimization (landmark training)	10	**5.0**
	M5.2	Privacy in detection phase	5	
	M5.3	Privacy in inference pipeline	0	
	M5.4	K-Anonymity considerations	5	
P6: Scientific Excellence	M6.1	Model comparison (word recognition)	10	**10.0**
	M6.2	Model comparison (hand detection)	10	
	M6.3	Statistical validation	10	
	M6.4	Literature grounding	10	
	M6.5	Honest reporting of limitations	10	
P7: Principled Availability	M7.1	Intended use clarity	10	**6.7**
	M7.2	Misuse potential assessment	5	
	M7.3	Open research contribution	5	
		Overall Average Score		**7.6**

Table 7.1: Ethical evaluation metrics based on Google's AI Principles [14]

7.5 Conclusions of the Ethical Analysis

The overall ethical evaluation of this project yields an average score of **7.6 out of 10** across all seven principles of Google's AI framework. This indicates that the project demonstrates a solid ethical foundation, with particularly strong results in scientific excellence (10.0/10) and safety (8.8/10), while also identifying clear areas for improvement, especially in privacy (5.0/10).

The strongest aspect of the project from an ethical perspective is the choice of using MediaPipe hand landmarks as the input representation for word recognition. This single architectural decision has positive implications across multiple principles: it reduces appearance based bias (Principle 2), it

acts as a data minimization mechanism that inherently protects privacy by discarding identifiable visual information (Principle 5), and it limits the potential for misuse since the numerical landmark data cannot be repurposed for surveillance or identification (Principle 7).

However, the evaluation also reveals that the privacy advantages of the landmark approach do not extend to all stages of the project. The hand detection phase was trained on real photographs of people from multiple datasets, and more importantly, the current server inference pipeline receives and stores full video frames without encryption or automatic deletion. This represents the most significant ethical gap identified in the analysis, and it should be the primary focus of improvement before any deployment beyond the research context. A recommended mitigation would be to move the landmark extraction to the client device (the mobile phone), so that only abstract numerical data is sent to the server, eliminating the need to transmit or store identifiable images.

The areas of bias avoidance (6.3/10) and principled availability (6.7/10) receive intermediate scores, reflecting the fact that while the project has inherent design choices that are positive for these dimensions, it lacks some formal processes. For bias, no demographic audit of the dataset signers was performed, and the hand detection datasets have limited or undocumented diversity. For availability, the absence of access controls and data retention policies on the server session storage is a gap that needs to be addressed.

Finally, it is worth noting that the honest reporting of limitations throughout the thesis including the significant accuracy drops at larger vocabularies, the documented instability above 700 classes, and the privacy weaknesses of the inference pipeline identified in this very chapter is itself an ethical practice aligned with Principle 6 (scientific excellence). Overestimating the capabilities of an AI system can lead to inappropriate deployment and user trust issues, so transparency about what the system can and cannot do is fundamental to responsible AI development.

Chapter 8

My Journey at UFV

Chapter 9

Conclusiones

9.1 General Conclusions

9.2 Conclusions regarding Project Objectives

9.3 Next Steps

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